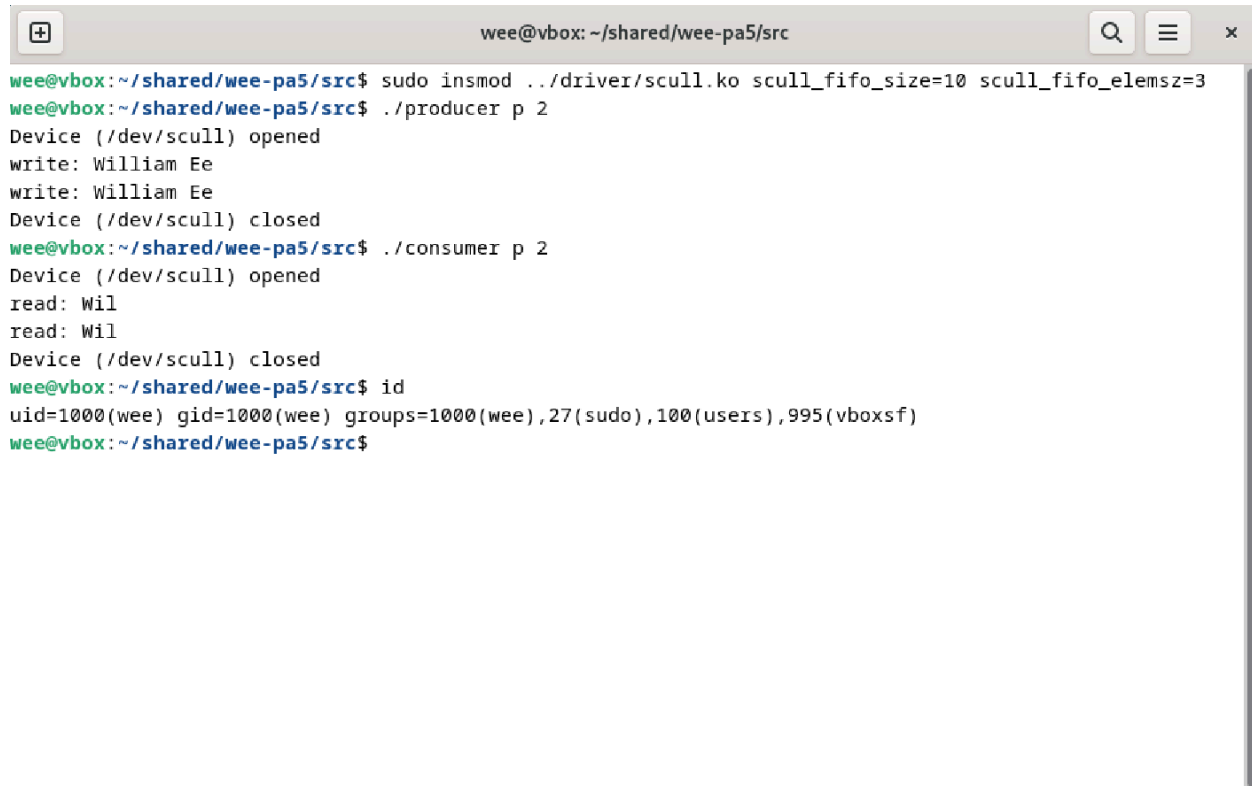


William Ee

Due 4/17/2025

I pledge my honor that I have abided by the Stevens Honor System.

5.1 - Small elemsz (3), many slots (10)

A terminal window titled 'wee@vbox: ~/shared/wee-pa5/src' with search, menu, and close icons. It shows the execution of a producer-consumer program using the scull kernel module. The producer writes 'William Ee' twice, and the consumer reads 'Wil' twice. The terminal output is as follows:

```
wee@vbox:~/shared/wee-pa5/src$ sudo insmod ../driver/scull.ko scull_fifo_size=10 scull_fifo_elemsz=3
wee@vbox:~/shared/wee-pa5/src$ ./producer p 2
Device (/dev/scull) opened
write: William Ee
write: William Ee
Device (/dev/scull) closed
wee@vbox:~/shared/wee-pa5/src$ ./consumer p 2
Device (/dev/scull) opened
read: Wil
read: Wil
Device (/dev/scull) closed
wee@vbox:~/shared/wee-pa5/src$ id
uid=1000(wee) gid=1000(wee) groups=1000(wee),27(sudo),100(users),995(vboxsf)
wee@vbox:~/shared/wee-pa5/src$
```

5.2 - Large elemsz (30), many slots (10)

```
wee@vbox: ~/shared/wee-pa5/src
wee@vbox:~/shared/wee-pa5/src$ sudo insmod ../driver/scull.ko scull_fifo_size=10 scull_fifo_elemsz=30
wee@vbox:~/shared/wee-pa5/src$ ./producer p 2
Device (/dev/scull) opened
write: William Ee
write: William Ee
Device (/dev/scull) closed
wee@vbox:~/shared/wee-pa5/src$ ./consumer p 2
Device (/dev/scull) opened
read: William Ee
read: William Ee
Device (/dev/scull) closed
wee@vbox:~/shared/wee-pa5/src$ id
uid=1000(wee) gid=1000(wee) groups=1000(wee),27(sudo),100(users),995(vboxsf)
wee@vbox:~/shared/wee-pa5/src$
```

5.3 - Killing producers / consumers

```
wee@vbox: ~/shared/wee-pa5/src
wee@vbox:~/shared/wee-pa5/src$ sudo insmod ../driver/scull.ko scull_fifo_size=2 scull_fifo_elemsz=30
wee@vbox:~/shared/wee-pa5/src$ ./producer p 5
Device (/dev/scull) opened
write: William Ee
write: William Ee
write: William Ee
write: William Ee
write: William Ee
^C
wee@vbox:~/shared/wee-pa5/src$ ./consumer p 5
Device (/dev/scull) opened
read: William Ee
read: William Ee
^C
wee@vbox:~/shared/wee-pa5/src$ id
uid=1000(wee) gid=1000(wee) groups=1000(wee),27(sudo),100(users),995(vboxsf)
wee@vbox:~/shared/wee-pa5/src$
```

5.4 - Run in two terminals

```
wee@vbox: ~/shared/wee-pa5/src
wee@vbox:~/shared/wee-pa5/src$ sudo insmod ../driver/scull.ko scull_fifo_size=2 scull_fifo_elemsz=30
wee@vbox:~/shared/wee-pa5/src$ ./producer p 5
Device (/dev/scull) opened
write: William Ee
write: William Ee
write: William Ee
write: William Ee
write: William Ee
Device (/dev/scull) closed
wee@vbox:~/shared/wee-pa5/src$

wee@vbox: /mnt/shared/wee-pa5/src
wee@vbox:/mnt/shared/wee-pa5/src$ ./consumer p 5
Device (/dev/scull) opened
read: William Ee
read: William Ee
read: William Ee
read: William Ee
read: William Ee
Device (/dev/scull) closed
wee@vbox:/mnt/shared/wee-pa5/src$ id
uid=1000(wee) gid=1000(wee) groups=1000(wee),27(sudo),100(users),995(vboxsf)
wee@vbox:/mnt/shared/wee-pa5/src$
```